Zopim F.C.Tablets 10 mg “S.T.”
(Zolpidem)

INGREDIENT
Each tablet contains:
Zolpidem Hemitartrate…………………………………10 mg

DESCRIPTION
It is a white, oblong, biconvex tablet, a S|T is scored on the one side and 385 on the other side.

CLINICAL PHARMACOLOGY
Pharmacodynamics
Subunit modulation of the $GABA_a$ receptor chloride channel macromolecular complex is hypothesized to be responsible for sedative, anticonvulsant, anxiolytic, and myorelaxant drug properties. The major modulatory site of the $GABA_a$ receptor complex is located on its alpha subunit and is referred to as the benzodiazepine (BZ) or receptor. At least three subtypes of the receptor have been identified.

While zolpidem is a hypnotic agent with a chemical structure unrelated to benzodiazepines, barbiturates, or other drugs with known hypnotic properties, it interacts with a GABA-BZ receptor complex and shares some of the pharmacological properties of the benzodiazepines. In contrast to the benzodiazepines, which non-selectively bind to and activate all three omega receptor subtypes, zolpidem *in vitro* binds the receptor preferentially. The receptor is found primarily on the Lamina IV of the sensorimotor cortical regions, substantia nigra (pars reticulata), cerebellum molecular layer, olfactory bulb, ventral thalamic complex, pons, inferior colliculus, and globus pallidus. This selective binding of zolpidem on the receptor is not absolute, but it may explain the relative absence of myorelaxant and anticonvulsant effects in animal studies as well as the preservation of deep sleep (stages 3 and 4) in human studies of zolpidem at hypnotic doses.

Pharmacokinetics
The pharmacokinetic profile of Zolpidem Hemitartrate is characterized by rapid absorption from the GI tract and a short elimination half-life ($T_{1/2}$) in healthy subjects. In a single- dose crossover study in 45 healthy subjects administered 5 and 10 mg Zolpidem Hemitartrate tablets, the mean peak concentrations (C_{max}) were 59 (range: 29 to 113) and 121 (range: 58 to 272) ng/ml, respectively, occurring at a mean time (T_{max}) of 1.6 hours for both. The mean Zolpidem Hemitartrate elimination half-life was 2.6 (range: 1.4 to 4.5) and 2.5 (range: 1.4 to 3.8) hours, for the 5 and 10 mg tablets, respectively. Zolpidem Hemitartrate is converted to inactive metabolites that are eliminated primarily by renal excretion. Zolpidem Hemitartrate demonstrated linear kinetics in the dose range of 5 to 20 mg. Total protein binding was found to be 92.5±0.1% and remained constant, independent of concentration between 40 and 790 ng/ml. Zolpidem did not accumulate in young adults following nightly dosing with 20 mg Zolpidem Hemitartrate tablets for 2 weeks.

A food-effect study in 30 healthy male volunteers compared the pharmacokinetics of Zolpidem Hemitartrate 10 mg when administered while fasting or 20 minutes after a meal. Results demonstrated that with food, mean AUC and C_{max}were decreased by 15% and 25% respectively, while mean $T_{max}$was prolonged by 60% (from 1.4 to 2.2 hr). The half-life remained unchanged. These results suggest that, for faster sleep onset, Zolpidem Hemitartrate should not be administered with or immediately after a meal.

In the elderly, the dose for Zolpidem Hemitartrate should be 5 mg. This recommendation is based on several studies in which the mean C_{max}, $T_{1/2}$, and AUC were significantly increased when compared to results in young adults. In one study of eight elderly subjects (> 70 years), the means for C_{max}, $T_{1/2}$, and AUC significantly increased by 50% (255 vs 284 ng/ml), 32% (2.2 vs 2.9 hr), and 64% (955 vs 1,562 ng · hr/ml), respectively, as compared to younger adults (20 to 40 years) following a single 20 mg oral Zolpidem dose. Zolpidem Hemitartrate did not accumulate in elderly subjects following nightly oral dosing of 10 mg for 1 week.

INDICATIONS
Insomnia

DOSAGE AND ADMINISTRATION
Dose - The treatment should always be implemented at the lowest effective dose and maximum dosage never exceeded.
The usual dose for adults is one 10mg tablet daily.
The medicinal product should always be taken just before going to bed.
In elderly subjects or subjects presenting with hepatic insufficiency, dosage should be halved, i.e. 5mg.
Dosage must never exceeded 10mg per day.
In the absence of clinical studies, Zolpidem should not be given to children.
Zolpidem can be prescribed either continuously or on demand, depending on the patient's symptoms.
Treatment duration - The treatment period should be as short as possible, from a few days to four weeks, including the tapering period.
The patients should be advised to take the treatment as follows :
2 to 5 days for occasional insomnia (e.g. during a trip)
2 to 3 weeks for transient insomnia (e.g. during a troubled period)
Very short treatment periods do not require any gradual treatment discontinuation.
For certain patients, it may be necessary to continue treatment for longer than four weeks, in which case careful and repeated reassessment of the patient's condition is necessary.

Mode of Administration
Oral

WARNING & PRECAUTIONS
ANAPHYLAXIS (SEVERE ALLERGIC REACTION) AND ANGIOEDEMA (SEVERE FACIAL SWELLING) WHICH CAN OCCUR AS EARLY AS THE FIRST TIME THE PRODUCT IS TAKEN.
COMPLEX SLEEP-RELATED BEHAVIORS WHICH MAY INCLUDE SLEEP DRIVING, MAKING PHONE CALLS, PREPARING AND EATING FOOD (WHILE ASLEEP).

The cause of insomnia should be identified wherever possible and the underlying factors treated before a hypnotic is prescribed. The failure of insomnia to remit after a 7-14 day course of treatment may indicate the presence of a primary psychiatric or physical disorder, and the patient should be carefully re-evaluated at regular intervals.

Specific patient groups
Elderly: See dose recommendations.
Depression: As with other sedative/hypnotic drugs, zolpidem tartrate should be administered with caution in patients exhibiting symptoms of depression. Suicidal tendencies may be present therefore the least amount of drug that is feasible should be supplied to these patients because of the possibility of intentional overdosage by the patient.
Use in patients with a history of drug or alcohol abuse: Extreme caution should be exercised when prescribing for patients with a history of drug or alcohol abuse. These patients should be under careful surveillance when receiving zolpidem tartrate or any other hypnotic, since they are at risk of habituation and psychological dependence.

General information
General information relating to effects seen following administration of benzodiazepines and other hypnotic agents which should be taken into account by the prescribing physician are described below.
Tolerance
Some loss of efficacy to the hypnotic effects of short-acting benzodiazepines and benzodiazepine-like agents may develop after repeated use for a few weeks.
Dependence
Use of benzodiazepines or benzodiazepine-like agents may lead to the development of physical and psychological dependence. The risk of dependence increases with dose and duration of treatment; it is also greater in patients with a history of psychiatric disorders and/or alcohol or drug abuse.
These patients should be under careful surveillance when receiving hypnotics.
Once physical dependence has developed, abrupt termination of treatment will be accompanied by withdrawal symptoms. These may consist of headaches or muscle pain, extreme anxiety and tension, restlessness, confusion and irritability. In severe cases the following symptoms may occur: derealisation, depersonalisation, hyperacusis, numbness and tingling of the extremities, hypersensitivity to light, noise and physical contact, hallucinations or epileptic seizures.
Rebound insomnia
A transient syndrome whereby the symptoms that led to treatment with a benzodiazepine or benzodiazepine-like agent recur in an enhanced form, may occur on withdrawal of hypnotic treatment. It may be accompanied by other reactions including mood changes, anxiety and restlessness.
It is important that the patient should be aware of the possibility of rebound phenomena, thereby minimising anxiety over such symptoms should they occur when the medicinal product is discontinued. Since the risk of withdrawal phenomena or rebound has been shown to be greater after abrupt discontinuation of treatment, it is recommended that the dosage is decreased gradually where clinically appropriate.
There are indications that, in the case of benzodiazepines and benzodiazepine-like agents with a short duration of action, withdrawal phenomena can become manifest within the dosage interval, especially when the dosage is high.
Amnesia
Benzodiazepines or benzodiazepine-like agents may induce anterograde amnesia. The condition occurs most often several hours after ingesting the product and therefore to reduce the risk patients should ensure that they will be able to have an uninterrupted sleep of 7-8 hours.
Psychiatric and "paradoxical" reactions
Reactions like restlessness, aggravated insomnia, agitation, irritability, aggressiveness, delusion, rages, nightmares, hallucinations, psychoses,

inappropriate behaviour and other adverse behavioural effects are known to occur when using benzodiazepines or benzodiazepine-like agents. Should this occur, use of the product should be discontinued. These reactions are more likely to occur in the elderly.

Pregnancy and Lactation

Pregnancy Category B

Teratogenic Effects: Studies to assess the effects of zolpidem on human reproduction and development have not been conducted.

Nonteratogenic Effects : Studies to assess the effects on children whose mother took zolpidem during pregnancy have not been conducted. However, children born of mothers taking sedative/hypnotic drugs may be at some risk for withdrawal symptoms from the drug during the postnatal period. In addition, neonatal flaccidity has been reported in infants born of mothers who received sedative/hypnotic drugs during pregnancy.

Nursing Mothers

Studies in lactating mothers indicate that the half-life of zolpidem is similar to that in young normal volunteers (2.6 ± 0.3 hr). Between 0.004 and 0.019% of the total administered dose is excreted into milk, but the effect of zolpidem on the infant is unknown.

The use of zolpidem tartrate in nursing mothers is not recommended.

Pediatric Use

Safety and effectiveness in children below the age of 18 have not been established.

Effect on ability to Drive and Use Machine

Zolpidem has major influence on the ability to drive and use machines.

Vehicle drivers and machine operators should be warned that, as with other hypnotics, there may be a possible risk of drowsiness, prolonged reaction time, dizziness, sleepiness, blurred/double vision and reduced alertness and impaired driving the morning after therapy. In order to minimize this risk a resting period of at least 8 hours is recommended between taking zolpidem and driving, using machinery and working at heights.

Driving ability impairment and behaviors such as 'sleep-driving' have occurred with zolpidem alone at therapeutic doses.

Furthermore, co-administration of zolpidem with alcohol and other CNS depressants increases the risk of such behaviors. Patients should be warned not to use alcohol or other psychoactive substances when taking zolpidem.

SIDE EFFECTS

There is evidence of a dose-relationship for adverse effects associated with zolpidem tartrate use, particularly for certain CNS and gastrointestinal events. As recommended in Recommended dose, they should in theory be less if zolpidem tartrate is taken immediately before retiring, or in bed. They occur most frequently in elderly patients.

Daytime drowsiness, reduced alertness, confusion, fatigue, headache, dizziness, muscle weakness, gait disturbances or diplopia. These phenomena usually occur predominantly at the start of therapy.

Other effects like gastrointestinal disturbances, changes in libido or skin reactions have been reported occasionally.

Amnesia – anterograde amnesia may occur using therapeutic dosages, the risk increasing at higher dosages. Amnestic effects may be associated with inappropriate behaviour.

Psychiatric and paradoxical reactions - reactions like restlessness, agitation, irritability, aggressiveness, delusion, rages, nightmares, hallucinations, psychoses, inappropriate behaviour, somnambulism and other adverse behavioural effects are known to occur when using zolpidem tartrate. Such reactions are more likely to occur in the elderly.

Depression – pre-existing depression may be unmasked during use of zolpidem tartrate. Since insomnia may be a symptom of depression, the patient should be re-evaluated if insomnia persists.

DRUG INTERACTIONS

CNS-Active Drugs

Zolpidem Hemitartrate was evaluated in healthy volunteers in single-dose interaction studies for several CNS drugs. A study involving haloperidol and zolpidem revealed no effect of haloperidol on the pharmacokinetics or pharmacodynamics of zolpidem. Imipramine in combination with zolpidem produced no pharmacokinetic interaction other than a 20% decrease in peak levels of imipramine, but there was an additive effect of decreased alertness. Similarly, chlorpromazine in combination with zolpidem produced no pharmacokinetic interaction, but there was an additive effect of decreased alertness and psychomotor performance. The lack of a drug interaction following single-dose administration does not predict a lack following chronic administration.

An additive effect on psychomotor performance between alcohol and zolpidem was demonstrated.

A single-dose interaction study with zolpidem 10 mg and fluoxetine 20 mg at steady-state levels in male volunteers did not demonstrate any clinically significant pharmacokinetic or pharmacodynamic interactions. When multiple doses of zolpidem and fluoxetine at steady-state concentrations were evaluated in healthy females, the only significant change was a 17% increase in the zolpidem half-life. There was no evidence of an additive effect in psychomotor performance.

Following five consecutive nightly doses of zolpidem 10 mg in the presence of sertraline 50 mg (17 consecutive daily doses, at 7:00 am, in healthy female volunteers), zolpidem C_{max} was significantly higher (43%) and T_{max} was significantly decreased (53%). Pharmacokinetics of sertraline and N-desmethylsertraline were unaffected by zolpidem.

Since the systemic evaluation of Zolpidem Hemitartrate in combination with other CNS-active drugs have been limited, careful consideration should be given to the pharmacology of any CNS-active drug to be used with zolpidem. Any drug with CNS-depressant effects could potentially enhance the CNS- depressant effects of zolpidem.

Drugs that affect drug metabolism via cytochrome P450: A randomized, double-blind, crossover interaction study in ten healthy volunteers between itraconazole (200 mg once daily for 4 days) and a single dose of zolpidem (10 mg) given 5 hours after the last dose of itraconazole resulted in a 34% increase in $AUC_{0\text{->}\infty}$ of zolpidem. There were no significant pharmacodynamic effects of zolpidem on subjective drowsiness, postural sway, or psychomotor performance.

A randomized, placebo-controlled, crossover interaction study in eight healthy female volunteers between 5 consecutive daily doses of rifampin (600 mg) and a single dose of zolpidem (20 mg) given 17 hours after the last dose of rifampin showed significant reductions of the AUC (–73%), C_{max} (–58%), and $T_{1/2}$ (–36%) of zolpidem together with significant reductions in the pharmacodynamic effects of zolpidem.

OVERDOSE

Signs and Symptoms

In European postmarketing reports of overdose with zolpidem alone, impairment of consciousness has ranged from somnolence to light coma. There was one case each of cardiovascular and respiratory compromise. Individuals have fully recovered from Zolpidem Hemitartrate overdoses up to 400 mg (40 times the maximum recommended dose). Overdose cases involving multiple CNS-depressant agents, including zolpidem, have resulted in more severe symptomatology, including fatal outcomes.

Recommended Treatment

General symptomatic and supportive measures should be used along with immediate gastric lavage where appropriate. Intravenous fluids should be administered as needed. Flumazenil may be useful. As in all cases of drug overdose, respiration, pulse, blood pressure, and other appropriate signs should be monitored and general supportive measures employed. Hypotension and CNS depression should be monitored and treated by appropriate medical intervention. Sedating drugs should be withheld following zolpidem overdosage, even if excitation occurs. The value of dialysis in the treatment of overdosage has not been determined, although hemodialysis studies in patients with renal failure receiving therapeutic doses have demonstrated that zolpidem is not dialyzable.

Poison Control Center

As with the management of all overdosage, the possibility of multiple drug ingestion should be considered. The physician may wish to consider contacting a poison control center for up- to-date information on the management of hypnotic drug product overdosage.

CONTRAINDICATIONS

Zolpidem Hemitartrate is contraindicated in patients with a hypersensitivity to Zolpidem Hemitartrate, obstructive sleep apnoea, myasthenia gravis, severe hepatic insufficiency, acute and/or severe respiratory depression. In the absence of data, Zolpidem Hemitartrate should not be prescribed for children or patients with psychotic illness.

SHELF LIFE:

3 Years

STORAGE CONDITION:

Keep this medication in the container it came in, tightly closed, and out of reach of children. Store it below 30℃ and away from excess heat and moisture (not in the bathroom).

PACKAGE:

10 x 10 tablets in PVC/foil blister strip / box

REGISTRATION NO.: MAL06091401AZ

Manufacturer:

TAIWAN BIOTECH CO., LTD.

22, Chieh-Shou Road, Taoyuan, Taiwan, R.O.C.

Product Registration Holder::

SUNWARD PHARMACEUTICAL SDN BHD

No. 9, 11 & 17, Jalan Kempas 4, Taman Perindustrian Tampoi Indah,
81200 Johor Bahru, Johor, Malaysia.

Revised Date: 28 Mar 2023